

SAP S/4HANA MM — Procure-to-Pay (P2P) Cycle

End-to-End Documentation

Düsseldorf Manufacturing AG (DMAG) — Fictional Sandbox Project

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Disclaimer: Fictional sandbox implementation. DMAG is a fictional company used solely to illustrate SAP MM P2P configuration.

This document presents a complete SAP S/4HANA MM Procure-to-Pay (P2P) cycle executed end-to-end for a manufacturing scenario, demonstrating my practical capability in procurement execution, pricing, GR/IR, invoice verification, and FI-MM integration. It highlights my ability to configure, execute, analyse, and troubleshoot P2P processes, including RFQ, quotation comparison, PO creation, goods receipt, invoice verification, and L2/L3 issue resolution, reflecting real-world procurement operations in the DACH region.

1. P2P Overview

The Procure-to-Pay cycle is the end-to-end process that begins when a business need is identified (Purchase Requisition) and ends when the vendor is paid (Invoice Verification). This document covers the full cycle executed for DMAG, including standard P2P, outline agreements, special scenarios and troubleshooting.

Scenario Context

Parameter	Value
Requesting Plant	DM02 — Dortmund Production
Purchase Organization	DMPO — Central Purchasing, DMAG
Material	200001 — Steel Sheet (ROH)
Quantity Required	1,000 KG
Competing Vendors	VEND-DM01 Steel Supplies GmbH VEND-DM02 Aluminium Werke AG
Pricing Schema	ZDM01 — Gross Price + Discount + Freight

2. Step-by-Step P2P Execution

Step 1 — Purchase Requisition (PR)

PR raised by production planner for 1,000 KG Steel Sheet. Account assignment category K (cost center DM02-PROD).

Configuration	Execution / Data
- PR document type NB — number range, field selection configured - Account assignment categories K and P enabled - Mixed account assignment — cost split across cost objects	Material: 200001 Steel Sheet Qty: 1,000 KG Plant: DM02 Storage Loc: SL01 Acct Assignment: K — Cost Centre DM02-PROD T-Code: ME51N

Step 2 — Request for Quotation (RFQ)

Two RFQs created from PR via ME41 — one for each competing vendor.

Configuration	Execution / Data
- RFQ document type AN — dedicated number range configured - Pricing procedure ZDM01 active on quotation - Collective number assigned for multi-vendor comparison in ME49	Vendor 1: VEND-DM01 Steel Supplies GmbH Vendor 2: VEND-DM02 Aluminium Werke AG Collective No: DM2026-001 Deadline: 5 working days T-Code: ME41

Step 3 — Maintain Quotation

Vendor prices entered via ME47. Pricing procedure ZDM01 calculates net effective price.

Vendor	Gross Price	Discount	Freight	Net Total (EUR)
VEND-DM01 Steel Supplies GmbH	2.50 EUR/KG	5% (>500 KG)	100 EUR flat	2,475.00
VEND-DM02 Aluminium Werke AG	2.70 EUR/KG	None	100 EUR flat	2,800.00

T-Code: ME47 | Validity: 30 days | Condition analysis visible on each quotation line item.

Step 4 — Quotation Comparison

ME49 price comparison executed for collective number DM2024-001. VEND-DM01 was selected as lowest effective net price. VEND-DM02 marked for rejection.

- VEND-DM01 selected: 2,475.00 EUR — saving 325.00 EUR vs VEND-DM02
- Rejection reason code 01 applied to VEND-DM02 — rejection letter output triggered
- T-Code: ME49

Step 5 — Purchase Info Record

- Info Record created for VEND-DM01 and material 200001 at 2.50 EUR/KG
- Planned delivery time: 10 days | Validity: 12 months
- Delivery tolerance: 10% over, 10% under purchasing value key
- Source List activated — VEND-DM01 set as fixed source for MRP auto-PO via ME59N
- T-Code: ME11

Step 6 — Purchase Order (PO)

PO created for 1,000 KG Steel Sheet from VEND-DM01. Price auto-populated from Info Record. Net value 2,475 EUR — below 50,000 EUR release threshold.

PO Field	Value
Document Type	NB — Standard Purchase Order
Vendor	VEND-DM01 Steel Supplies GmbH
Material / Qty	200001 Steel Sheet 1,000 KG 2.50 EUR/KG
Net Value	2,475.00 EUR (gross 2,500 – discount 125 + freight 100)
Delivery	Plant DM02 SL01 Delivery date confirmed
GR-Based IV	Active — MIRO blocked until GR posted
Release Required	No — value 2,475 EUR below 50,000 EUR threshold
T-Code	ME21N

Step 7 — Goods Receipt (GR)

980 KG received (within 10% under-delivery tolerance). GR posted to DM02/SL01. Batch DM-BATCH-001 created with Grade-B classification.

GR Field	Value
Movement Type	101 — Goods Receipt for Purchase Order
Quantity	980 KG — within tolerance (1,000 KG ordered)
Stock Update	DM02 / SL01 Unrestricted Use: +980 KG
FI Posting Dr	130000 Inventory Account (BSX): 2,450.00 EUR
FI Posting Cr	191100 GR/IR Clearing (WRX): 2,450.00 EUR
T-Code	MIGO

Step 8 — Invoice Verification (MIRO)

Vendor invoice for 980 KG at 2.50 EUR/KG = 2,450 EUR. Three-way match: PO price, GR quantity and invoice amount all match. GR/IR clearing account cleared.

MIRO Field	Value
Invoice Amount	2,450.00 EUR Vendor: VEND-DM01
Three-Way Match	PO price match GR quantity match Invoice amount match
FI Posting Dr	191100 GR/IR Clearing (WRX): 2,450.00 EUR — cleared
FI Posting Cr	Vendor Account VEND-DM01: 2,450.00 EUR — AP liability created
T-Code	MIRO

3. Additional P2P Scenarios

Scenario	Configuration	Key T-Codes
Value Contract	12-month framework, 500,000 EUR total. Item cat W (material group). Release orders drawn against contract.	ME31K / ME21N / ME3M
Scheduling Agreement	Periodic delivery for Aluminum Rod across 4 months. Type LP. Delivery schedule via ME38.	ME31L / ME38 / MIGO
Auto Purchase Order	Info Record and Source List active. ME59N converts approved PRs to POs automatically.	ME01 / ME11 / ME59N
Parked Invoice	Invoice saved without posting — pending approval. Reviewed and posted via MIR6.	MIRO / MIR6
GR Without PO	Unplanned receipt, movement type 501. Account assignment required at GR.	MIGO — 501
Unplanned Del Costs	Additional freight added at MIRO. Allocated to material cost or separate GL account.	MIRO
Quota Arrangement	10,000 KG split 60/40 between VEND-DM01 and VEND-DM02. Auto-PO assigns proportionally.	MEQ1 / ME01 / ME59N

4. L2/L3 Troubleshooting

Issue	Root Cause	Resolution Steps
GR/IR Mismatch	Invoice posted for different qty or value than GR. Or GR reversed without corresponding invoice reversal.	1. Run MB5S to identify open GR/IR items 2. Check PO history for GR and invoice quantities 3. Post credit memo in MIRO or reverse incorrect GR 4. Clear residual items via MR11
PO Release Block	Release characteristic values on PO do not match configured strategy. Approver missing authorization.	5. Check release status in ME29N 6. Verify CEKKO field values match strategy conditions 7. Check user authorization for release code 8. Reset release and re-trigger if classification incorrect
Pricing Error on PO	Wrong condition value on PO. Info Record price incorrect or access sequence not finding correct table.	9. Check condition analysis in PO (Analysis button) 10. Verify Info Record price via ME13 11. Check access sequence ZDM1 for correct table hit 12. Update condition record via MEK1 and re-determine price

MIRO Invoice Block	Invoice posted but payment blocked. Price or quantity variance outside tolerance. Manual block set.	13. Check tolerance key settings in OMR6 14. Review MIRO document for variance details 15. Release block manually via MRBR if variance accepted 16. Request credit memo from vendor for price discrepancy
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5. FI-MM Integration Summary

Transaction	T-Code	Dr Account	Cr Account	OBYC Key
Goods Receipt	MIGO-101	130000 Inventory (BSX)	191100 GR/IR Clearing (WRX)	BSX/WRX
Invoice Posted	MIRO	191100 GR/IR Clearing (WRX)	Vendor Account (AP)	WRX
Price Difference	MIRO	281000 Price Variance (PRD)	Vendor Account (AP)	PRD
Goods Issue	MIGO-261	400000 Consumption (GBB-VBR)	130000 Inventory (BSX)	GBB/BSX
Return to Vendor	MIGO-122	191100 GR/IR Clearing (WRX)	130000 Inventory (BSX)	BSX/WRX

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